

BioShield: A Tree Bark-Inspired Self-Adaptive Fire Protection Materials for Safer Buildings

1 Abstract (100 words)

Fire remains one of the greatest threats to modern buildings, causing severe human, economic, and environmental consequences. Current fire protection systems are often passive and unable to adapt to changing fire conditions. Inspired by the natural survival strategies of trees, which protect themselves from extreme heat through layered, porous bark structures [1], this report proposes BioShield: a self-adaptive fire protection material combining lightweight insulation, heat-resistant structures, and temperature sensing capabilities. By mimicking biological thermal defence mechanisms [2], BioShield aims to provide improved fire resistance, early hazard detection, and sustainable protection for future buildings while reducing dependence on heavy conventional fireproof materials.

2 Problem

Building fires continue to represent a significant global safety challenge, with increasing urbanisation, high-rise construction, and the use of advanced synthetic materials increasing fire complexity. Despite advances in fire safety engineering, current passive fire protection systems such as concrete barriers, mineral wool, and intumescent coatings remain largely static in function [3]. They are designed to withstand predefined fire scenarios but cannot actively respond to variations in temperature, flame exposure, or changing environmental conditions.

The limitations of existing fire protection materials create several engineering challenges. Traditional fire-resistant materials often require significant thickness and mass, increasing structural loading and construction costs. Furthermore, many protective coatings degrade after exposure to high temperatures and provide limited information about their condition during or after a fire event. A next-generation fire protection system should therefore combine thermal resistance, structural durability, lightweight design, sustainability, and the ability to respond to fire development.

The objective of this project is to develop a bio-inspired fire protection material capable of delaying heat transfer and maintaining structural integrity under extreme temperatures. Therefore, a next-generation fire protection material should therefore achieve:

- a) Thermal conductivity below approximately 0.05 W/mK to provide effective insulation.
- b) Fire resistance capable of protecting structural elements for 60–120 minutes.
- c) Lightweight construction with a target density below 300 kg/m³.
- d) Ability to withstand thermal shock and repeated heating cycles.
- e) Integration of temperature sensing capability for fire monitoring.
- f) Use of sustainable and non-toxic materials where possible.

Several constraints must also be considered. Any fire protection material used in buildings must satisfy strict safety regulations, including fire resistance testing, smoke toxicity requirements, and structural performance standards [3]. The material must not release harmful gases during combustion or degradation. Manufacturing cost and scalability are also important because large quantities of fire protection materials are required in construction applications. Additionally, long-term durability under humidity, mechanical loading, and environmental exposure must be demonstrated before commercial implementation.

3 Inspiration

The biological inspiration for BioShield comes primarily from the protective structure of tree bark, particularly species such as cork oak (*Quercus suber*), which survive extreme environmental conditions including wildfire exposure. Tree bark functions as a multilayered thermal protection system that sacrifices its outer layers during fire exposure while protecting inner living tissues [1]. The cellular structure of bark contains numerous air-filled pores, reducing heat transfer through trapped air and providing lightweight insulation.

This biological strategy is inspiring because it demonstrates that effective thermal protection does not necessarily require dense, heavy materials. Instead, protection can be achieved through hierarchical structures combining sacrificial layers, porous insulation, and mechanical reinforcement.

A second source of inspiration is biological skin. Human skin provides both protection and sensing functionality by combining a protective outer layer with embedded sensory systems capable of detecting temperature and damage. This principle suggests that future fire protection materials should not only resist fire but also monitor their own condition.

Finally, nacre (mother-of-pearl) provides a model for mechanical reinforcement. Its layered microstructure combines hard and soft phases, improving toughness and preventing crack propagation under stress [2], [4].

Together, these biological systems provide three key engineering principles:

- 1) Tree bark → layered thermal protection and porous insulation [1].
- 2) Skin → integrated sensing and response capability.
- 3) Nacre → hierarchical structures for mechanical durability [4].

These principles form the foundation of the BioShield concept.

4 The Solution

BioShield is proposed as a multilayer composite fire protection panel designed for integration into building walls, structural elements, and fire compartments. The proposed structure consists of four functional layers shown in Figure 4.1, which was drawn by using Canvas.

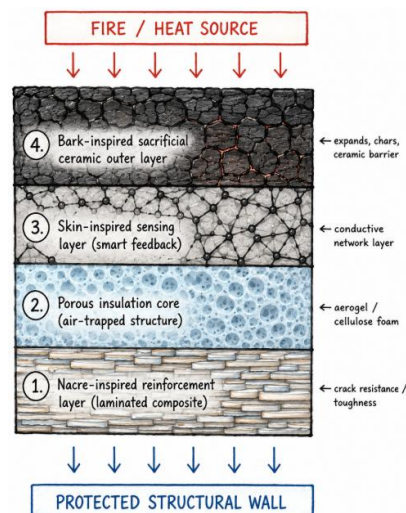


Figure 4.1: Schematic representation of the BioShield multilayer fire protection panel showing functional stratification inspired by bark, plant tissue, skin, and nacre.

4.1 Bark-inspired sacrificial ceramic outer layer

The external layer mimics tree bark by forming a protective char/ceramic barrier when exposed to high temperatures. A ceramic-polymer composite containing materials such as alumina nanoparticles and expandable graphite would undergo controlled expansion and transformation during heating. This creates a low-density protective layer that reduces flame penetration and slows heat transfer.

As shown in Figure 4.2, the process mirrors natural bark-like protective behaviour under thermal exposure [1].

In stage 1 the material starts as a smooth, dense layer that responds to initial heat input by activating internal protective mechanisms. This early response is critical, as the structure begins to stabilise rather than immediately breaking down, setting the foundation for later insulation. In the next stage, continued heating drives gas release within the material, causing it to expand into a porous char foam structure. This expansion builds the thermal resistance, as seen in the diagram where the solid block becomes increasingly cellular. By stage 3, extreme temperatures gradually consume the material, leaving behind a porous ceramic residue that still provides thermal shielding. This final state reflects the controlled, sacrificial breakdown, where protection is maintained even as material is progressively consumed.

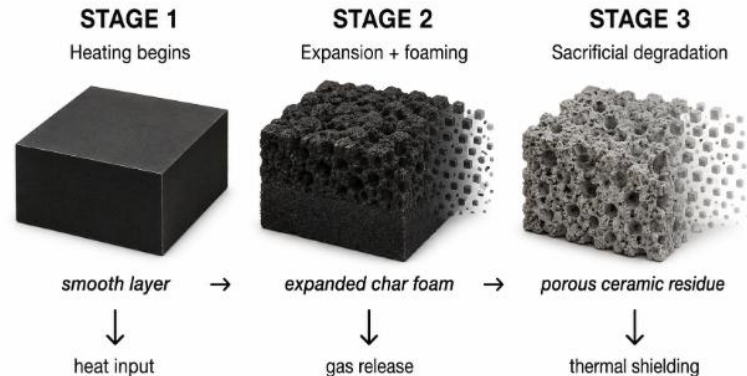


Figure 4.2: Thermally induced transformation of the bark-inspired outer layer demonstrating expansion, char formation, and sacrificial degradation under extreme exposure.

4.2 Porous insulation core inspired by plant cellular structures

The middle layer is designed as a lightweight, porous structure inspired by plant tissues and can be implemented as a bio-inspired aerogel composite incorporating cellulose-derived materials (shown in figure 4.3) [2]. The trapped air within the interconnected porous network significantly reduces thermal conductivity, since air is a poor conductor of heat, a strategy that closely mimics natural insulating systems such as wood and bark. The target properties for this layer include a density of less than 300 kg/m^3 , a thermal conductivity below 0.05 W/mK , and high temperature stability.

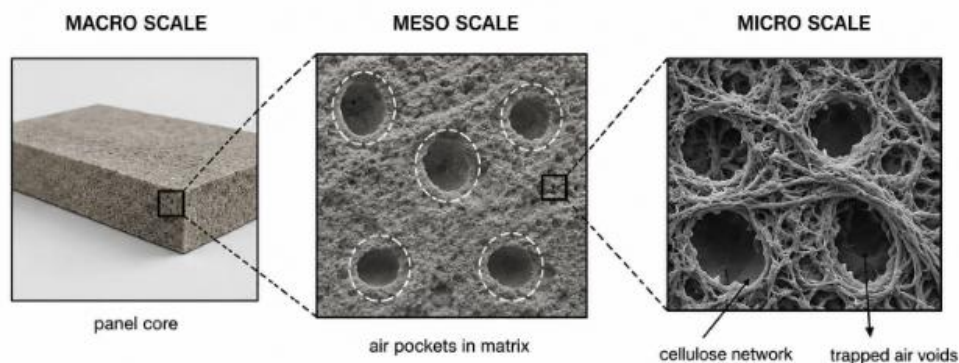


Figure 4.3: Hierarchical porous architecture of the plant-inspired insulation core.

4.3 Skin-inspired sensing layer

A major innovation of BioShield is the integration of a sensing function, where a thin embedded conductive network monitors temperature changes through variations in electrical resistance.

As shown in Figure 4.4, during a fire the temperature increase leads to a measurable change in electrical resistance, which in turn triggers a warning signal, effectively converting the material's thermal response into an electrical output. This transforms BioShield from a purely passive fire barrier into a “smart” system capable of communicating its condition in real time rather than simply resisting heat.

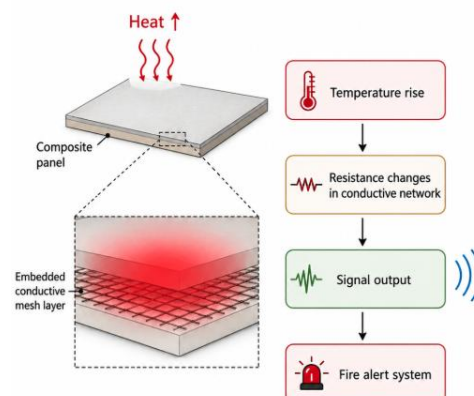


Figure 4.4: Conceptual representation of the integrated skin-inspired sensing layer.

4.4 Nacre-inspired reinforcement layer

The BioShield reinforcement layer is inspired by the hierarchical “brick-and-mortar” structure of nacre (mother-of-pearl), which achieves high toughness despite being made from relatively brittle materials. This architecture is used to improve the mechanical reliability of the composite under thermal and mechanical stress.

The layer consists of alternating hard and soft phases. The hard phase is formed by ceramic particles such as alumina or silica, which provide stiffness, compressive strength, and thermal stability, acting as rigid structural “bricks”. The soft phase is a polymer matrix that acts as a compliant “mortar”, absorbing stress and allowing controlled deformation between ceramic regions.

This combined structure improves performance through damage-control mechanisms rather than relying only on material strength. Under loading, cracks are slowed and redirected at the interfaces between layers (crack deflection), while the polymer phase helps maintain partial load transfer across cracks (crack bridging) [4][5]. In addition, the soft phase redistributes stress caused by thermal expansion mismatch, reducing stress concentrations that would otherwise lead to fracture.

Layer	Bio Inspiration	Material Concept	Primary Function	Key Mechanism
Outer sacrificial layer	Tree bark	Ceramic–polymer composite (Al_2O_3 + expandable graphite)	Fire shielding & thermal barrier formation	Expansion + char/ceramic formation
Sensing layer	Human skin	Conductive polymer / resistive network	Fire detection & monitoring	Resistance change with temperature
Insulation core	Plant tissue	Cellulose-based aerogel	Heat insulation	Air trapping & low thermal conductivity
Reinforcement layer	Nacre (mother-of-pearl)	Layered ceramic–polymer composite	Structural stability	Crack deflection & stress distribution

Table 4.1: Functional roles and biomimetic design principles of each BioShield layer.



5 Feasibility Considerations

Heat transfer through the material can be approximated using Fourier's law:

$$Q = \frac{kA\Delta T}{L}$$

where:

Q = heat transfer rate

k = thermal conductivity

A = surface area

ΔT = temperature difference

L = thickness

Reducing thermal conductivity through porous structures and increasing effective thickness reduces heat flow into the protected structure.

For example, reducing thermal conductivity from conventional concrete (~1.5 W/mK) to an aerogel-inspired layer (~0.05 W/mK) could theoretically reduce conductive heat transfer by approximately 97%, assuming equal geometry.

Further development would require:

- a. Furnace fire testing.
- b. Cone calorimeter testing.
- c. Mechanical compression testing.
- d. Thermal cycling experiments.
- e. Computational modelling of heat propagation.

6 Evaluation

BioShield provides several advantages compared with conventional fire protection systems. Its main strength is that it combines multiple biological strategies into one multifunctional material [1][2]. Traditional materials usually provide only insulation, whereas BioShield integrates thermal resistance, structural reinforcement, and sensing capability.

Economically, the initial manufacturing cost would likely be higher than conventional insulation due to advanced composite fabrication and sensor integration. However, reduced structural weight, improved fire safety, and potential reduction in maintenance requirements may provide long-term economic benefits. Large-scale manufacturing methods would need development before commercial application.

Implementation presents several challenges. Fire protection materials must undergo extensive certification testing before use in buildings. The interaction between polymer components, ceramic layers, and sensing electronics under extreme fire conditions requires further investigation.

The solution meets the original specifications by providing:

1. Lightweight fire protection.
2. Improved thermal insulation.
3. Fire-response monitoring.
4. Sustainable material possibilities.

However, several limitations remain. The exact behaviour of the composite during prolonged fires is uncertain, and laboratory-scale performance may not directly translate to real building environments [3]. The durability of embedded sensors after repeated heating cycles also requires validation.

Safety and sustainability are important considerations. BioShield could incorporate renewable cellulose-based materials, reducing dependence on petroleum-based insulation. However, potential emissions from polymer degradation must be assessed carefully. Future research should focus on fully recyclable and non-toxic material combinations.

Overall, BioShield demonstrates how biological principles can inspire advanced engineering solutions for fire safety. While further research and testing are required, the concept provides a promising pathway towards adaptive, lightweight, and sustainable fire protection technology.



7 References

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